Sequences, Series and Ad-hoc

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I. Find the sum
$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\cdots+\frac{1}{\sqrt{n}+\sqrt{n+1}}.$$

II. Let $\{a_k\}$ be an arithmetic progression with common difference d. Compute
$$\sum_{k=1}^{n}\frac{1}{a_k a_{k+1} a_{k+2}}$$

III. Suppose that a sequence $a_1, a_2, \ldots$ of positive real numbers satisfies
$$a_{k+1} \geq \frac{k a_k}{a_k^2+(k-1)}$$
for every positive integer k. Prove that $a_1+a_2+\cdots+a_n \geq n$ for every $n \geq 2$.

IV. Let a be a real number not equal to $0, \pm\frac{1}{\sqrt{3}}$, or $\pm\sqrt{3}$, and define a sequence $x_1, x_2, x_3, \ldots$ by
$$x_{n+1}=\frac{\sqrt{3}x_n+1}{\sqrt{3}-x_n} \quad \text{for } n \geq 1.$$
Prove that this sequence is periodic, that is, there exists a positive integer k such that $x_{n+k}=x_n$ for all positive integers n.

V. A sequence of real numbers $a_0, a_1, \ldots$ is defined as follows. a_0 is an arbitrary real number and for $n \geq 0$, $a_{n+1}=\lfloor a_n \rfloor \{a_n\}$. (Here, $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x, and $\{x\}=x-\lfloor x \rfloor$). Prove that $a_n=a_{n+2}$ for n sufficiently large.

VI. Consider the sequence $x_1, x_2, x_3, \ldots$ defined by
$$x_1=a, \quad x_{n+1}=4x_n(1-x_n) \quad \text{for } n \geq 1.$$
Show that

(a) if $0 \leq a \leq 1$, then $0 \leq x_n \leq 1$ for all positive integers n;

(b) there exists $0 \leq a \leq 1$ such that $x_{2005}=x_1$, and $x_n \neq x_1$ for $2 \leq n \leq 2004$.

VII. Find a formula in compact form for the general term of the sequence defined recursively by $x_1=1$, $x_n=x_{n-1}+n$ if n is odd, and $x_n=x_{n-1}+n-1$ if n is even.

VIII. A sequence $(a_n)_{n\geq 0}$ satisfies $a_{m+n}+a_{m-n}=\frac{1}{2}(a_{2m}+a_{2n})$ for all integers m,n with $m \geq n \geq 0$. Given that $a_1=1$, find a_{2003}.

IX. Find the general term of the sequence defined by $x_0=3, x_1=4$ and
$$x_{n+1}=x_{n-1}^2-nx_n$$
for all $n \in \mathbb{N}$.

X. Prove that

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \dots \cdot \frac{2n-1}{2n} < \frac{1}{\sqrt{3n}}$$

for all positive integers n.

XI. A sequence of real numbers $a_1, a_2, \dots$ is called *interesting* if $a_1 > 0$ and, for every $n \geq 2$,

$$a_1 a_2 \dots a_{n-1} a_n = a_1 + a_2 + \dots + a_{n-1}.$$

What is the maximum number of integers that such a sequence can contain?
Note: the product has n factors and the sum has $n-1$ summands.

XII. Fix a positive integer n. Define sequences $a, b, c \in \mathbb{Q}^{n+1}$ by $(a_0, b_0, c_0) = (0, 0, 1)$ and

$$a_k = (n-k+1) \cdot c_{k-1}, \quad b_k = \binom{n}{k} - c_k - a_k, \quad \text{and} \quad c_k = \frac{b_{k-1}}{k}$$

for each integer $1 \leq k \leq n$.
Determine for which n it happens that $a, b, c \in \mathbb{Z}^{n+1}$.